

MAT31-5

Applied Probability, Simulation, and Experimental Inference

Monte Carlo, variance reduction, bootstrapping, A/B testing, likelihood geometry

Estimating quantities by simulation, quantifying their uncertainty, and turning both into decisions.

Albert School — 22 hours

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1 Monte Carlo estimation

Many quantities of interest can be written as an expectation

$$\mu = \mathbb{E}[g(X)],$$

where X is a random variable (or vector) with a known distribution and g is a function. Examples: a probability is $\mathbb{E}[\mathbb{1}_A(X)]$, a mean payoff is $\mathbb{E}[g(X)]$, an integral $\int g(x)f(x) dx$ is $\mathbb{E}[g(X)]$ when f is the density of X .

Definition 1 (Sample average approximation) Draw independent samples $X_1, \dots, X_n$ from the distribution of X and set

$$\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n g(X_i).$$

By the Law of Large Numbers (MAT31-2), $\hat{\mu}_n \rightarrow \mu$ as $n \rightarrow \infty$.

Proposition 2 (Standard error) Let $\sigma^2 = \text{Var}(g(X))$ and write $\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (g(X_i) - \hat{\mu}_n)^2$ for the sample variance. The estimator $\hat{\mu}_n$ is unbiased with

$$\text{Var}(\hat{\mu}_n) = \frac{\sigma^2}{n}, \quad \text{SE}(\hat{\mu}_n) = \frac{\sigma}{\sqrt{n}} \approx \frac{\hat{\sigma}}{\sqrt{n}}.$$

The error decreases like $1/\sqrt{n}$: dividing the standard error by two requires four times as many samples.

Definition 3 (Monte Carlo confidence interval) By the Central Limit Theorem (MAT31-2), for large n an approximate $(1 - \alpha)$ confidence interval for μ is

$$\hat{\mu}_n \pm z_{1-\alpha/2} \frac{\hat{\sigma}}{\sqrt{n}},$$

where $z_{1-\alpha/2}$ is the standard normal quantile (e.g. 1.96 for 95%).

2 Variance reduction

The width of the Monte Carlo confidence interval is proportional to $\text{SE}(\hat{\mu}_n) = \sigma/\sqrt{n}$. **Variance-reduction** techniques replace the naive estimator by another unbiased estimator of the same μ with smaller variance, narrowing the interval at fixed n .

Definition 4 (Antithetic variates) Suppose $g(X)$ can be evaluated on a sample X and on a paired sample X' with the same distribution but negatively correlated outputs (e.g. $X = F^{-1}(U)$ and $X' = F^{-1}(1 - U)$ for U uniform on $[0, 1]$). With $n/2$ such pairs, use

$$\hat{\mu}^{\text{anti}} = \frac{1}{n/2} \sum_{i=1}^{n/2} \frac{g(X_i) + g(X'_i)}{2}.$$

Its variance is

$$\text{Var}(\hat{\mu}^{\text{anti}}) = \frac{\sigma^2}{n} (1 + \rho), \quad \rho = \text{Corr}(g(X), g(X')),$$

so a negative correlation $\rho < 0$ reduces variance relative to σ^2/n .

Definition 5 (Control variates) Let C be a variable, computed alongside $g(X)$, whose mean $\mathbb{E}[C] = c$ is known. For a constant b ,

$$\hat{\mu}^{\text{cv}} = \frac{1}{n} \sum_{i=1}^n (g(X_i) - b(C_i - c))$$

is unbiased for μ . Its variance is minimised at $b^* = \text{Cov}(g(X), C) / \text{Var}(C)$, giving

$$\text{Var}(\hat{\mu}^{\text{cv}}) = \frac{\sigma^2}{n} (1 - \rho_{g,C}^2),$$

where $\rho_{g,C}$ is the correlation between $g(X)$ and C . The stronger the correlation, the larger the reduction.

To **quantify the impact** of a strategy, compute the standard error and the confidence-interval width with and without it: the ratio of variances equals $1 + \rho$ (antithetic) or $1 - \rho_{g,C}^2$ (control variates), and the interval width scales as the square root of that ratio.

3 Bootstrapping

Monte Carlo assumes a known distribution to simulate from. When all we have is a **finite sample** $x_1, \dots, x_n$ from an unknown distribution, the **bootstrap** treats that sample as the population and resamples from it.

Definition 6 (Bootstrap) Let $\hat{\theta} = T(x_1, \dots, x_n)$ be a statistic. For $b = 1, \dots, B$: draw a **resample** $x_1^*, \dots, x_n^*$ by sampling n values with replacement from the original data, and compute $\hat{\theta}_b^* = T(x_1^*, \dots, x_n^*)$. The spread of the values $\hat{\theta}_1^*, \dots, \hat{\theta}_B^*$ estimates the sampling variability of $\hat{\theta}$.

Definition 7 (Bootstrap standard error and confidence interval) The **bootstrap standard error** is the sample standard deviation of the $\hat{\theta}_b^*$. The **percentile** $(1 - \alpha)$ confidence interval is the interval between the $\alpha/2$ and $1 - \alpha/2$ empirical quantiles of the $\hat{\theta}_b^*$.

Common pitfall The bootstrap can **fail**. It is unreliable for **small samples** (the resamples cannot contain information the original sample lacks) and for **heavy-tailed distributions** or statistics dominated by extremes (e.g. the maximum), where the resampling distribution does not approximate the true sampling distribution.

4 Simulation of stochastic systems

Some systems resist closed-form analytical treatment: queues, inventories, multi-stage random processes. They are studied by **model-based simulation**.

Definition 8 (Simulation workflow)

1. Specify the model: state variables, the random inputs and their distributions, and the rules that advance the system.
2. Generate the random inputs and run one **replication**, recording the outcome of interest.
3. Repeat for n independent replications to obtain $Y_1, \dots, Y_n$.
4. Report the mean outcome $\bar{Y}$ with a confidence interval $\bar{Y} \pm z_{1-\alpha/2} \hat{\sigma}_Y / \sqrt{n}$, exactly as in Monte Carlo estimation.

5 Validation

Simulation outputs must be checked before they are trusted.

- **Analytical benchmark.** When a special case has a closed-form answer, run the simulation on that case and verify the estimate agrees with the known value within its confidence interval.
- **Convergence diagnostics.** When no closed form is available, plot the running estimate $\hat{\mu}_n$ against n and check that it stabilises, and that the standard error shrinks like $1/\sqrt{n}$.

6 Python applications

Monte Carlo workflows are implemented in Python. A representative estimator with its standard error and 95% confidence interval:

```
import numpy as np

def monte_carlo(g, sampler, n, alpha=0.05):
    x = sampler(n)          # n samples of X
    y = g(x)                # g(X)
    mu = y.mean()
    se = y.std(ddof=1) / np.sqrt(n)  # standard error
    z = 1.959963985         # normal 0.975 quantile
    return mu, se, (mu - z*se, mu + z*se)
```

The same pattern computes replicated simulation outcomes, and the bootstrap is one resampling loop:

```
def bootstrap_ci(data, stat, B=10000, alpha=0.05):
    n = len(data)
    boot = np.array([stat(np.random.choice(data, n, replace=True))
```

```

        for _ in range(B)])
    return np.quantile(boot, [alpha/2, 1 - alpha/2])

```

Outputs are **validated** against an analytical benchmark or a convergence plot as above.

7 A/B testing

An **A/B test** is a controlled experiment comparing two variants (control A and treatment B) by randomly assigning units to each and comparing an outcome.

Definition 9 (Design) State the null and alternative hypotheses (e.g. $H_0 : p_A = p_B$ vs $H_1 : p_A \neq p_B$), the significance level α , the target power $1 - \beta$, and the **minimum detectable effect** δ deemed worth detecting.

Proposition 10 (Sample-size logic) For a two-proportion test at level α with power $1 - \beta$ to detect a difference δ , the per-group sample size is approximately

$$n \approx \left(z_{1-\alpha/2} + z_{1-\beta} \right)^2 \frac{2\bar{p}(1-\bar{p})}{\delta^2},$$

where $\bar{p}$ is the assumed baseline rate. Smaller δ , smaller α , or larger power all **increase** the required n .

Common pitfall (Multiple testing) Running many tests inflates the chance of a false positive: at level α , the probability that at least one of m independent true nulls is rejected is $1 - (1 - \alpha)^m$. Correct for this, e.g. by the **Bonferroni** rule (test each at level α/m).

Common pitfall (Stopping rule) Repeatedly checking the test and stopping as soon as it is significant (**peeking**) also inflates the Type I error rate. Fix the sample size in advance, or use a method designed for sequential monitoring.

8 Statistical vs. practical significance

Definition 11 A result is **statistically significant** when the data are unlikely under H_0 (small p-value). It is **practically significant** when the estimated effect is large enough to matter for the decision at hand.

With a large enough sample, a negligible effect can be statistically significant; with a small sample, an important effect can fail to reach significance. A decision must weigh the **effect size** and its confidence interval — i.e. **decision relevance** — not the p-value alone.

9 Likelihood geometry

For a model with two parameters $\theta = (\theta_1, \theta_2)$ and data, the **log-likelihood** $\ell(\theta)$ defines a surface over the parameter plane. We read it operationally, using the multivariable tools of MAT31-2 / MAT21-2, not deriving it from scratch.

Definition 12 (Gradient and Hessian) The **gradient** $\nabla\ell(\theta) = (\partial\ell/\partial\theta_1, \partial\ell/\partial\theta_2)$ points uphill; the maximum-likelihood estimate $\hat{\theta}$ is where $\nabla\ell = 0$. The **Hessian** $H(\theta)$ collects the second derivatives and measures the **curvature** at the peak.

At $\hat{\theta}$ the Hessian is negative definite (MAT31-1). Its curvature encodes parameter uncertainty:

- **Sharp** curvature (large $|H|$) $\rightarrow$ a narrow peak $\rightarrow$ small uncertainty: the parameter is well determined.
- **Flat** curvature $\rightarrow$ a broad peak $\rightarrow$ large uncertainty.
- A **ridge** — flat along one direction — signals **weak local identifiability**: the data constrain a combination of parameters but not each separately. This shows as elongated, tilted contour ellipses.

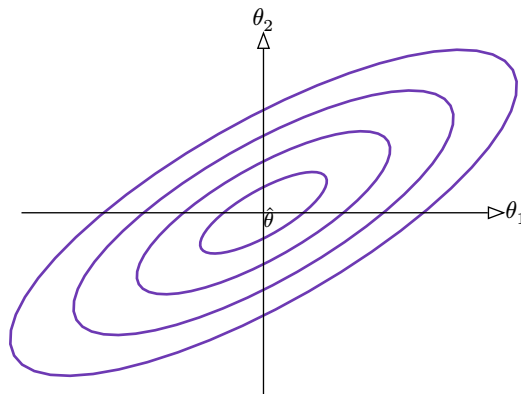


Figure 1: Contours of a two-parameter log-likelihood. Elongated, tilted ellipses indicate a ridge: large uncertainty along the long axis and weak local identifiability.

10 A complete simulation-and-inference workflow

Combining the above into one decision-oriented analysis:

1. **State the estimand** as an expectation $\mu = \mathbb{E}[g(X)]$ or a system outcome.
2. **Choose a simulation strategy**: direct Monte Carlo, or replicated simulation of a stochastic system.
3. **Apply one variance-reduction technique** (antithetic or control variates) and report its effect on the standard error.
4. **Compute an inferential summary**: the estimate with a confidence interval (Monte Carlo CLT interval or bootstrap interval).
5. **Validate** against an analytical benchmark or convergence diagnostics.
6. **Write a decision memo**: the estimate, its uncertainty, statistical vs. practical significance, stated limitations, and the validation checks performed.